Project Management & Engineering (PME)
Ma'aden Engineering Manual

Title: Water Treatment Plant

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REVISION INDEX

Rev	Date	Description
C01	15 Dec 2011	Issued for Use
C02	12 Feb 2018	This Spec has been re-written on the following basis using information from the Projects and Vendor documentation. Generally WTP consists of package units including pressure vessels, filled with resin for the specific treatment required. The Vessels have changeover valves for back flushing and regeneration of the resign. Chemical dosing units provide the reagents for the regeneration and treatment to meet the process requirements. Reverse Osmosis units have been added as they are a major component in certain applications. The Spec includes references to current Project Specs for the equipment to meet Project discipline requirements. Other Specifications used to draft this Spec are listed below. MD-510-0000-EG-ME-SPC-0078 – Specification for Reverse Osmosis Plant. MD-510-0000-EG-ME-SPC-0004 – Packaged and Fabricated Plant (EPC) MD-510-0000-EG-ME-SPC-0032 – Chemical Dosing Package MD-510-0000-EG-ME-SPC-0036 - Specification for metering pumps MD-510-0000-EG-ME-SPC-0044 - Specification for FRP vessels and tanks MD-510-0000-EG-ME-SPC-0064 - Specification for Water Treatment Package MD-510-0000-EG-ME-SPC-0066 – Side Stream Filter Appendices have been added to illustrate the typical Water Treatment parameters that are expected. These would be conformed depending on the required application.

Rev.	Rev. Date	Description	Prepared By	Checked By	Approved By	Employer
C02	09 Feb 2018	Issued for Use	AB		FYG	
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1. INTRODUCTION

1.1 Scope

This specification outlines the requirements of the Water Treatment Plant (WTP) for Ma'aden (Saudi Arabian Mining Company) located in the Kingdom of Saudi Arabia.

It is not the intent to specify all details of design and construction; nevertheless, the Plant Equipment shall conform in all respects to the highest standards of engineering, design and workmanship.

Compliance with this specification shall not relieve the Vendor of the responsibility of supplying Plant Equipment and auxiliaries of proper design, material, and workmanship to meet the operating requirements defined within this specification. In case of additional instruments, controls, safety devices and any other auxiliaries or accessories required for safe, smooth and trouble free operation of the Plant Equipment, the Vendor shall include the same in the scope of supply and provide reasons in the offer for the addition.

The Vendor may also provide an alternative process design / technology, Plant Equipment arrangement, cycle times, etc. including explanations of the benefits of the alternative(s).

2. KEY TERMS AND DEFINITIONS

The following is a list of the principal abbreviations employed within the body of this Specification

AISC	American Institute of Steel Construction
API	American Petroleum Institute
ASD	Allowable Strength Design
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
BFD	Block Flow Diagram
CIP	Cleaning In Place
dB (A)	Decibel (A Weighted Scale)
DCS	Distributed Control System
FRP	Fiber Reinforced Plastic
GRP	Glass Reinforced Plastic
HAZOP	Hazard and Operability
ISO	International Standards of Organization
LRFD	Load and Resistance Factor Design
MCC	Motor Control Centre
MTBF	Mean Time Between Failures

NDT	Non Destructive Testing
NEMA	National Electrical Manufacturers Association
PFD	Process Flow Diagram
P&ID	Piping and Instrumentation Diagram
PTP	Pre-treatment packages
RAM	Reliability, Availability & Maintainability
RO	Reverse Osmosis
SIL	Safety Integrity Level
SWL	Safe Working Load
UF	Ultrafiltration
WTP	Water Treatment Plant

3. APPLICABLE DOCUMENTS

All work performed and all materials furnished under this Specification shall comply with the applicable statutory requirements of the Kingdom of Saudi Arabia, the requirements of this specification, and the standard specifications, drawings and equipment data sheets as referenced on the cover sheet of this Specification, and the latest revision of Codes, Standards and Regulations referenced herein

3.1 Statutory Regulations

All work performed and all materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia (including but not limited to directives of the High Commission for Industrial Safety and Security, Royal Commission regulations applicable to the Site, regulations of the Presidency of Meteorology and Environment, Saudi Building Code and other applicable codes and regulations). The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, by-laws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes And Standards

All work performed and all materials furnished under this Specification shall comply with the requirements of the codes and standards listed below.

3.2.1 ISO Standards

ISO 281	Rolling Bearings - Dynamic Load Ratings and Rating Life
ISO 1940-1	Mechanical Vibration – Balance Quality Requirements of Rigid Rotors – Part 1: Determination of Permissible Residual Unbalance

ISO 5199	Technical Specification for Centrifugal Pumps – Class II
ISO 9906	Roto-dynamic Pumps - Hydraulic Performance Acceptance Tests
ISO 10816-3	Mechanical vibration – Part 3: Evaluation of machine vibration by measurements on non-rotating parts.
ISO 14122	Safety of machinery **—** Permanent means of access to machinery
ISO 14691	Petroleum and Natural Gas Industries – Flexible Coupling for Mechanical Power Transmission – General Purpose Applications

3.2.2 ASME / ASTM Standards

ASME RTP-1	Reinforced Thermoset Plastic Corrosion-Resistant Equipment
ASME B16.11	Forged Fittings, Socket-Welding and Threaded
ASME B16.5	Pipe Flanges and Flanges Fittings
ASME B73.1	Specification for Horizontal End Suction Centrifugal Pumps for Chemical Process
ASME B73.2	Specification for Vertical In-Line Centrifugal Pumps for Chemical Process
ASME Sec VIII Div I & II	Rules for Construction of Pressure Vessels
ASME Section IX	Welder's Qualification
ASME Section X	Fibre-Reinforced Plastic Pressure Vessels
ASTM A36M	Standard Specification for Carbon Structural Steel
ASTM E 114	Ultrasonic Testing
ASTM A153/A153M	Standard Specification for Zinc Coated (Hot-Dip) on Iron and Steel Hardware
ASTM A193/A193M	Standard Specification for Alloy-Steel and Stainless Steel Bolting for High Temperature or High Pressure Service and Other Special Purpose Applications
ASTM A194/A194M	Standard Specification for Carbon Steel and Alloy Steel Nuts for Bolts for High Pressure or High Temperature Service
ASTM B633	Standard Specification for Electro-deposited Coatings of Zinc on Iron and Steel
ASTM F1554	Standard Specification for Anchor Bolts, Steel, 36, 55 and 105-KSI Yield Strength
ASTM D4516	Standard Practice for Standardizing Reverse Osmosis Performance Data

3.2.3 API Standards

API 620	Design &Construction of Large, Welded, Low-Pressure Storage Tanks
API 650	Atmospheric Storage Tanks
API 686	Recommended Practice for Machinery Installation and Installation Design
API 614	Lubrication, Shaft sealing and Control oil systems and Auxiliaries 5th Ed
API 617	Axial & Centrifugal Compressors & Expanders Compressors for Petroleum, Chemical and Gas Industry Services 7th Ed
API 618	Reciprocating Compressors for Petroleum, Chemical and Gas Industry Services 4th Ed
API 619	Rotary Type Positive Displacement Compressors for Petroleum, Chemical and Gas Industry Services 3th Ed
API 672	Packaged, Internally geard Centrifugal Compressors for Petroleum, Chemical and Gas Industry

3.2.4 Other Standards

ASCE 7	Minimum Design Loads for Buildings and Other Structures
BS EN 10204	Metallic Products ~ Type of inspection document
BS EN 10241	Steel Threaded Pipe Fittings
IBC	International Building Code
IEC 60034	Rotating Electrical Machines
MSS SP-55	Quality Standard for Steel Castings for Valves, Flanges and Fittings and Other Piping Components – Visual Method for Evaluation of Surface Irregularities
SBC 201	Saudi Building Code - Architectural
SBC 301	Saudi Building Code – Structural – Loading and Forces

3.2.5 Ma'aden Standards

MD-101-SMEM-EG-GE-CRT-0001	Plant Layout Design Criteria
MD-101-SMEM-EG-EL-CRT-0001	Electrical Design Criteria
MD-101-SMEM-EG-EL-SPC-0001	Electrical And Instrumentation Installation
MD-101-SMEM-EG-EL-SPC-0004	HV & LV Electric Motors
MD-101-SMEM-EG-EL-SPC-0007	HV and LV Variable Speed Drives
MD-101-SMEM-EG-EL-SPC-0013	Prefabricated Switch rooms
MD-101-SMEM-EG-GE-SPC-0002	Welding Specification

MD-101-SMEM-EG-HS-CRT-0001	Environmental, Health & Safety Design Criteria
MD-101-SMEM-EG-IN-SPC-0001	Instrumentation
MD-101-SMEM-EG-IN-SPC-0002	Control Valves Spec.
MD-101-SMEM-EG-ME-CRT-0001	Design Criteria - Mechanical - Stationary Equipment
MD-101-SMEM-EG-ME-CRT-0002	Mechanical Design Criteria - Rotating Equipment
MD-101-SMEM-EG-ME-SPC-0002	Mechanical Equipment Installation Specification
MD-101-SMEM-EG-ME-SPC-0004	Pressure Vessels
MD-101-SMEM-EG-ME-SPC-0006	Insulation for Cold Equipment & Piping
MD-101-SMEM-EG-ME-SPC-0007	Protective Coating
MD-101-SMEM-EG-ME-SPC-0009	Insulation of Hot Equipment & Piping
MD-101-SMEM-EG-ME-SPC-0013	Metallic Storage Tanks
MD-101-SMEM-EG-ME-SPC-0014	Centrifugal Chemical Pumps
MD-101-SMEM-EG-ME-SPC-0015	Positive Displacement Pumps
MD-101-SMEM-EG-PI-CRT-0001	Piping Design Criteria
MD-101-SMEM-EG-PI-SPC-0001	Piping Flexibility and Stress Analysis Specification
MD-101-SMEM-EG-PI-SPC-0002	Piping Material Specification
MD-101-SMEM-EG-PI-SPC-0004	Specification for Manual Valves
MD-101-SMEM-EG-PI-SPC-0005	Piping Installation and Testing
MD-101-SMEM-EG-ST-SPC-0008	Protective Coatings for Concrete
MD-101-SMEM-EG-ST-SPC-0009	Protective Coatings for Steelwork
MD-101-SMEM-EG-CV-SPC-0002	Anchor Bolts
MD-101-SMEM-EG-CV-SPC-0007	Geo Membranes

3.3 Alternative Codes and Standards

Alternative Codes and Standards may be used if approved by the Employer.

4. DESIGN REQUIREMENTS

4.1 General

The reliability of the WTP and its ability to run without un-planned failures, is of key importance and a mandatory requirement. The MTBF for major components shall be at

least 25,000 operating hrs. Auxiliary Plant Equipment such as CIP pumps, fans and filters shall be provided with installed spares.

Transmitters shall be employed rather than switches for alarm / trip functions. In principle, single component failure shall not cause a shutdown of the WTP. The Vendor shall describe the design features of the WTP which will minimize operator intervention and give good operational reliability.

Required availability of the WTP and its ability to run for extended periods without planned maintenance is considered a key advantage. The Vendor shall define the time required for all planned maintenance activities over a five year period.

Routine maintenance requirements shall be minimized where possible. For activities such as lubrication of pump and motor bearings consideration shall be given to alternative synthetic lubricants where these can provide extended intervals and increased reliability.

Vendor shall define the recommended spare parts holding and need for special tools in order to minimize the time for planned maintenance.

WTP shall be subject to model reviews, HAZOP and SIL reviews. The Vendor shall attend all reviews and implement agreed recommendations into the WTP design.

Vendor shall optimize his design such that the WTP can easily be maintained. Where common Plant Equipment is employed an installed spare shall be included.

Package shall be designed to maximize factory assembly and testing, thus simplifying and reducing to a minimum installation and commissioning operations. Modules are preferred to reduce the work on Site.

Package shall be wired and designed to ensure minimum installation operations on site. All external connections shall be brought to skid edge. Segregation shall be as per Project Specification MD-510-0000-EG-EL-SPC-0007 “Electrical Requirements for Packaged Equipment”.

Any structure or item not installed prior to transportation shall be fully erected in the Vendor’s workshop and match-marked for ease of reassembly at site.

4.2 Design Life

Plant Equipment shall have a design life per Project requirements, with the primary objectives of Plant Equipment reliability and ease of maintenance.

4.3 Site Conditions And Utilities

Refer to Project Site Specification.

4.4 Safety And Environment

Refer to Project Specification MD-101-SMEM-EG-HS-CRT-0001 “Environmental, Health & Safety Design Criteria”.

4.5 Plant Layout

Vendor shall provide an optimized Plant Equipment arrangement with good access for maintenance and operation considering the ergonomics requirements specified in Project Specification MD-101-SMEM-EG-GE-CRT-0001 “Plant Layout Design Criteria”.

Layout and design shall include the following :

- Access for maintenance and for normal operations i.e. position of instruments, legibility of readouts, operation of valves, etc. If necessary, an access platform with ladder shall be provided for Plant Equipment that cannot be accessible from the ground.
- Removable gratings or plates shall be provided around the machines to allow access and to protect Plant Equipment located in the bottom part of the package.
- Maintainability (disassembly, reassembly, adjustment) with the necessary space and handling devices as appropriate to bring any piece of Plant Equipment to dedicated lay down areas.

The above requirements shall be fully supported by an engineering access and handling study during contract execution for the following components but not limited to: drivers, rotors, couplings, electric motors, gears, piping spools, etc.

4.6 Reliability, Availability And Maintainability (RAM)

Equipment shall have a design life as required by the Project, with the primary objectives of Equipment reliability and ease of maintenance.

The number of forced outages shall be minimized and high reliability shall be achieved by incorporating standardised component designs and materials. No prototype components shall be considered. The selected Equipment shall be within the Vendor's proven experience and shall not involve the use of any prototype design or components.

Equipment shall be designed with the need for access, inspection, cleaning and repair either in situ or by removal to the workshop. Equipment shall incorporate features to reduce maintenance down time such as ease of component replacement or adjustment of individual components.

5. PLANT REQUIREMENTS

5.1 Pre-Treatment Packages

PTP shall comprise of all required Plant Equipment with all necessary tanks, pumps, compressors/blowers, chemical dosing skids and other ancillaries as required.

Pre-treatment is required for the reverse osmosis (RO) plant to ensure fouling of RO membranes is not excessive and thereby leads to excessive cleaning frequency of the RO membranes).

Vendor is to consider the following systems:

- Decarbonation
- Sodium hypochlorite dosing
- Caustic soda dosing
- Media filtration using a sand media
- Ultrafiltration (UF) using pressurized “out to in” flow configuration UF membrane modules
- Pumping upstream of pressurized media filters. Discharge pump pressure to be sufficient to pump water through pressurized sand filters and UF to the RO Feed water pumps under all operating conditions.
- Various water quality and flow instrumentation

5.2 RO Package

Vendor shall provide a reverse osmosis system complete with membranes, housings, control system, prefilters, piping and instrumentation within the battery limits and shall include all required instrumentation and controls for automatic operation. Vendor furnished PLC shall communicate with Client DCS system.

Where feasible, equipment shall be shipped skid mounted and pre-piped and pre-wired. All power and Instrument / Control Wiring between equipment / skids shall be run on fitted cable trays to Junction Boxes mounted on the Skid Frame.

Vendor shall provide equipment, tanks shop fabricated to the largest extent feasible for means of transport. Bolted segments shall be shop assembled; piece marked and knocked down for shipment.

The largest component shipped shall be less than or equal to 3.8 meters wide by 3 meters high by 12 meters in length. The design shall include platforms, stairs, ladders, and gates, including grating, handrails and toe plate.

Package control shall preferably be through the Employer's DCS system. However, the Vendor may provide an alternative proposal for the PLC and shall make the I/O available for supervisory control and monitoring to the main DCS.

All critical functions, alarming and troubleshooting shall be sent to the Owner's DCS for remote monitoring.

Vendor is to consider the following systems:

- Acid Dosing System
- Inline Static Mixer
- Antiscalant Injection System
- Sediment Pre-filter
- High Pressure Pump
- Reverse Osmosis Membrane Unit
- Instruments and controls
- Piping and Valves
- Sample Ports
- Structural Steel Skid
- Clean in Place System (CIP)

5.3 Clean In Place (CIP)

Reverse osmosis package shall be provided with a clean-in-place system to circulate cleaning chemicals through the reverse osmosis membranes.

CIP System shall be complete with all necessary appurtenances, instruments and equipment as required to make a complete self-contained assembly.

CIP System shall be pre-wired and pre-piped and shall contain controls to provide for automatic operation with minimum operator attention.

Vendor is to consider the following systems:

- Chemical Solutions Tank
- Strainer
- Centrifugal Pump
- Prefilters
- Supply and Return Piping
- Instruments and Valves

5.4 Waste Water Treatment

Dirty filter wash water from the media filters and UF shall be disposed to a wastewater effluent pit, or per Project requirements.

Waste streams eg: CIP wastes from UF and RO plant shall be neutralized prior to discharge to the wastewater effluent pit, or per Project requirements.

Regenerate and other wastes from other plants eg: demineralization water treatment plant / condensate polishing plant shall be neutralized prior to discharge to the wastewater effluent pit, or per Project requirements.

5.5 Materials

Materials of construction shall be suitable and proven for the operating conditions of the plant.

Materials of construction used shall be selected on the basis of minimum capital expenditure consistent with maximum reliability as required by the Project. As such there should be no predicted downtime due to material corrosion or material performance failures.

5.6 Pressure Vessels

Pressure vessels shall be designed according to Project Specification MD-101-SMEM-EG-ME-SPC-0004 “Pressure Vessels” and MD-101-SMEM-EG-ME-SPC-0010 “Pressure Equipment - Supplemental Requirements”.

Materials shall be certified to meet code standards and shall be traceable for each specific vessel.

5.7 GRE / FRP Vessels And Tanks

GRE / FRP Vessels and tanks fabricated by the hand lay-up or contact molding methods shall be in accordance with ASME RTP 1.

GRE / FRP Vessels and tanks fabricated by the filament winding process shall be in accordance with ASME RTP 1 or equivalent standard. The reinforcing material shall be Type E glass meeting ASTM D578.

ASTM D2563 "Standard Practice for Classifying Visual Defects in Glass-Reinforced Plastic Laminate Parts" shall be used for quality control of both filament-wound and hand lay-up construction.

Applicable design code like ASME RTP-1 (Reinforced Thermoset Plastic) and other applicable technical specifications shall be made clear in the proposal.

All conflicts between the requirements of this specification, the standards referenced above, the data sheets, and the purchase order shall be referred to Owner for clarification before manufacture begins.

Hydrostatic test pressure shall be as per code.

5.8 Tanks

Tanks shall be in compliance with Project Specification MD-101-SMEM-EG-ME-SPC-0013 “Metallic Storage Tanks”.

5.9 Pumps

Vendor is to consider Manufacturers Standard Pump range, without modification or special requirements that increase cost and schedule.

Centrifugal pumps shall be in accordance with the Project Specification MD-101-SMEM-EG-ME-SPC-0014 “Centrifugal Chemical Pumps”.

Positive Displacement Pumps shall be in accordance with the Project Specification MD-101-SMEM-EG-ME-SPC-0015 “Positive Displacement Pumps“

Slurry Pumps shall be in accordance with the Project Specification MD-101-SMEM-EG-ME-SPC-0003 “Centrifugal Pumps for Slurry Service”

5.10 Blowers (Centrifugal and Positive Displacement)

Blowers shall comply to the specifications listed in section 3.0 of this specification.

5.11 Chemical Dosing

Dosing package shall be delivered in the form of a self-contained packaged unit, ready for bolting down on prepared foundations with a minimum of field construction.

Package shall be erected on a concrete foundation located at grade unless otherwise specified on the Plant Equipment data sheets.

Reasonable access shall be provided for operation and maintenance.

Necessary access and maintenance ladders and platforms shall be provided.

Base plate shall be supplied with the necessary sub plates, shims and leveling jack to allow satisfactory leveling.

5.11.1 Dosing Tanks / Pressure Vessels

Tanks containing chemicals hazardous to health shall not be directly vented to atmosphere, without the use of appropriate filters, seals or scrubbing devices.

Storage tanks and mixers shall be provided with sufficient maintenance and access requirements.

Tank shall be sized for a suitable capacity.

Dosing tank bottom drain nozzle shall be at least 450mm above the top of the skid steel for adequate access.

Tanks shall be provided with a level gauge and level transmitter for low and high level alarms. Manual gauging shall be provided.

All pressure vessels such as pulsation dampeners shall be designed according to the ASME code section VIII, Division 1 or equivalent.

5.11.2 Dosing Pumps

Dosing pumps shall be positive displacement diaphragm type.

Pumps stroke adjustment shall be automatic or manual as specified on the data sheets.

Pump shall be equipped with suction & discharge isolation valves, pressure gauge, and pressure safety valve.

Pressure safety valve shall be flanged.

Full flow external relief valve shall be provided on the discharge side of each pump.

Pulsation dampener shall be provided where required.

5.12 Standard Asset Tagging And Identification

Refer to Project Specification MD-101-SMEM-EG-GE-PRO-0003 "Standard Asset Tagging and Identification".

5.13 Rotational Arrows

Direction of rotation arrows shall be clearly and permanently marked on all rotating Plant Equipment.

5.14 Painting

Refer to Project Specification MD-101-SMEM-EG-ME-SPC-0007 "Protective Coating".

5.15 Insulation

Vendor shall install Heat Conservation/Personnel Protection Insulation and Cold Conservation/Personnel Protection Insulation as required.

The Vendor shall supply and fit thermal insulation and cladding (with vapour barrier for services which may fall below dew point). The insulation shall be in accordance with Project Specification MD-101-SMEM-EG-ME-SPC-0006 "Insulation of Hot Equipment & Piping" and MD-101-SMEM-EG-ME-SPC-0009 "Insulation for Cold Equipment and Piping Specification".

5.16 Noise

The maximum sound pressure level shall be as specified on the mechanical data sheet.

Noise limits shall be in accordance with MD-101-SMEM-EG-ME-CRT-0002 "Mechanical Design Criteria - Rotating Equipment" and MD-101-SMEM-EG-HS-CRT-0001 "Environmental Health & Safety Design Criteria".

5.17 Vibration

Vibration limits for rotating Equipment shall be in accordance with MD-101-SMEM-EG-ME-CRT-0002 "Mechanical Design Criteria - Rotating Equipment"
Anti-vibration mounts shall be provided when applicable.

6. PACKAGE BASE PLATE, STRUCTURE & STEELWORK

6.1 Steelwork

Steel structures shall comply with Royal Commission Guideline Specifications for Steel (Sections 05120, 05210 and 05310), and Employer's Specifications:

MD-101-SMEM-EG-ST-SPC-0001	"Steel Fabrication",
MD-101-SMEM-EG-ST-SPC-0002	"Steel Work Erection".
MD-101-SMEM-EG-ST-CRT-0001	"Design Criteria - Process Buildings & Structures "

6.2 Package Baseplate / Structure Construction

When installed on concrete foundations, the package base plate shall be a multipoint design with jacking bolts for leveling purposes.

Baseplate / structure shall be sufficiently rigid to maintain alignment of Plant Equipment under all transportation and operating conditions whether normal, transient and abnormal.

The package base plate shall be designed with either:

- A non slip decking steel plate covering the top of the base plate and a drip pan and gutter all around with two drain points of 2" diameter minimum, or
- A steel plate continuously welded at the bottom of the base plate recovering any spillage and drain points of 2" diameter minimum.

6.3 Ladders, Platforms, Stairs And Handrails

The Vendor shall provide all ladders, handrails, catwalks, stairs, etc. required for normal operation, maintenance, and access to the components and systems.

- Access shall be provided to all valves, instruments, sight glasses, man ways, etc.
- The design shall not restrict the use of Plant Equipment for lifting components for maintenance and repairs.

6.4 Foundation Bolts

Foundation bolts shall be as per Project Standard MD-101-SMEM-EG-CV-SPC-0002 "Anchor Bolts".

6.5 Grouting

Grouting shall be as per Project Standard MD-101-SMEM-EG-ST-SPC-0003 "Grout"

Vendor shall include alignment, surface preparation, form work, setting and finishing, and Quality inspections for all grouting required to complete the Package installation.

6.6 Package Lifting

Refer to Project Specification MD-101-SMEM-EG-ME-SPC-0002 "Mechanical Equipment Installation Specification".

Package lifting appliances, such as lifting beams or spreader bars, slings, shackles, etc., shall be included in the Vendor's scope of supply and designed, tested and certified by an approved authority.

Package lifting appliances shall be provided for transportation (loading & unloading) and installation at site. Where multiple identical packages are provided, a single lifting beam etc may be supplied provided that all packages are shipped together at the same time. Where phased shipments are foreseen then multiple lifting beams etc shall be provided to suit the number of shipments and / or logistics.

7. PIPING

Piping and tubing shall be designed, fabricated, tested and installed in accordance with Project Specification MD-101-SMEM-EG-PI-SPC-0002 "Piping Material Specification".

In general, all Plant Equipment piping systems shall have flanged high point vents and flanged low point drains and shall terminate with an isolating valve and a blind flange.

Skid interconnecting piping, including drains, vents, auxiliary fluids connections at skid edge, shall be identified and tagged.

Drains and vents shall be routed to the skid edge and shall terminate by a flange fitted with an isolating valve and a blind flange.

Unique connection devices shall be used for chemicals to mitigate the risk of incorrect connection.

Drains and vents for lube-oil systems shall be valved and routed back to each relevant reservoir or common header.

Single block and bleed valve assembly shall be installed in each instrument take-off connection, except thermo well and vibration probes, and shall be located as close to the parent pipe or Plant Equipment as possible.

Take-off connections, including the first valve shall be in accordance with the piping specification.

Threaded connections on any pipe work are not permitted. Mitered bends are not permitted.

8. PIPING ISOLATIONS

Piping isolations shall comply with MD-101-SMEM-EG-PI-SPC-0012 "Specification for Piping Isolation".

9. ELECTRICAL

Package electrical scope shall comply with Project Specification MD-101-SMEM-EG-EL-CRT-0001 "Electrical Design Criteria" and other documents referenced in Section3 "Applicable Documents"

10. INSTRUMENTATION & CONTROLS

Package instrumentation and control Plant Equipment shall comply with the following Project Specifications

- MD-101-SMEM-EG-CS-SPC-0001 Template for Process Control Requirements Specification

- MD-101-SMEM-EG-CS-SPC-0002 Process Control System - Hardware
- MD-101-SMEM-EG-CS-SPC-0003 Process Control System - Configuration / Design Criteria (CRT) - Instrumentation
- MD-101-SMEM-EG-CS-SPC-0004 Plant Control System - Network Equipment
- MD-101-SMEM-EG-CS-SPC-0005 Process Control System - Fiber Optic Cable Installation
- MD-101-SMEM-EG-CS-SPC-0006 Communication Design and Installation

WTP control shall preferably be through the Employer's DCS system. However, the Vendor may provide an alternative proposal for the PLC and shall make the I/O available for supervisory control and monitoring to the main DCS.

PLC must be capable of interfacing with the plant DCS and the SIS for operation and safeguarding, by suitable redundant serial communication and hardwired connections.

If a PLC is provided, a HMI shall be installed in the packaged equipment control panel to allow local operator interface.

Sensors, signal conditioners and associated racks / panels/ JBs are included in the package Vendor scope of supply.

Vendor shall also provide functional control narratives, Piping & Instrumentation Diagrams (P&ID's), Cause & Effect, and JB schedules etc. to the DCS for Contractor development of control and protection logic in the DCS as well as DCS wiring.

11. INSPECTION AND TESTING

Refer to Project Specification MD-101-SMPM-PM-QA-PEI-0001 "Contractor Quality Requirements" and MD-510-0000-QD-QA-SPC-1001 "Quality Program Requirements for Contractors"

12. SPARE PARTS AND SPECIAL TOOLS

Vendor shall furnish a unit priced list of recommended spare parts for start-up and two years operation. Spare parts shall be identified with the Vendor's part number and all information shall be provided on the Project Data Base Input Spread Sheet.

Major spare parts may be purchased with the Plant Equipment. Prices and recommended major spare parts shall be submitted to the Employer for review.

Special tools required for the maintenance of the plant Equipment shall be provided by the Vendor.

13. PREPARATION FOR SHIPMENT

Refer to Project Specification MD-101-SPPM-CP-LO-PEI-0002 "Shipping Requirements" and MD-101-SPPM-PO-LO-PEI-0004 "Warehousing Requirements".

Vendor shall provide Storage and Preservation procedures, and undertake the same to ensure that the Warranty is not affected.

APPENDIX A

NOMINAL WATER QUALITY REQUIREMENTS

(TO BE VERIFIED DURING DESIGN)

PRETREATED WATER QUALITY REQUIREMENT PRIOR TO RO

Parameter	Requirement
pH	6.0 to 7.5
Turbidity (NTU) downstream of UF	< 0.3 on a 99th percentile basis < 0.25 on a 95th percentile basis
Turbidity (NTU) downstream of Sand filters If UF treatment for colloidal silica removal NOT installed downstream.	< 1.0 NTU on a 95th percentile basis < 1.5 NTU on a maximum basis
Turbidity (NTU) downstream of Sand filters If UF treatment for colloidal silica removal NOT installed downstream.	< 0.2 NTU on a 95th percentile basis < 0.3 NTU on a maximum basis
Dissolved Organic Carbon (DOC)	< 2.0 mg/l
SDI15 measured downstream RO cartridge filtration	< 3.0 on a 99th percentile basis < 4.5 on a maximum basis
Free chlorine residual (mg/l) – measured at a point after cartridge filtration (CF) and Bi-sulphate dosing prior to RO membranes.	0.0
Oxidation Reduction Potential (ORP) measured after dechlorination prior to R.O. membrane feed (mV) membrane feed (mV)	< 290 mV or as required by RO membrane supplier
Total iron	<10 ug/l
Total Manganese	<10 ug/l

FIRST PASS RO PERMEATE QUALITY REQUIREMENTS

PARAMETER (mg/l) UNLESS OTHERISE NOTED (U.O.N)	TYPICAL QUALITY 1 YEAR OPERATION	TYPICAL QUALITY 5 YEAR OPERATION	CONTRACTOR GUARANTEED QUALITY (mg/l) U.O.N
TDS	Contractor To Confirm	Contractor To Confirm	< 100
Conductivity (us/cm)	Contractor To Confirm	Contractor To Confirm	< 105
Chloride	Contractor To Confirm	Contractor To Confirm	< 70
Silica	Contractor To Confirm	Contractor To Confirm	< 1
pH	Contractor To Confirm	Contractor To Confirm	Contractor to advise based on relevant acid and anti scalant usage
Calcium	Contractor To Confirm	Contractor To Confirm	< 2
Magnesium	Contractor To Confirm	Contractor To Confirm	< 1
Total Hardness (mg/l as CaCO3)	Contractor To Confirm	Contractor To Confirm	< 5
Fluoride	Contractor To Confirm	Contractor To Confirm	< 0.1
Sodium	Contractor To Confirm	Contractor To Confirm	< 45
Sulphate	Contractor To Confirm	Contractor To Confirm	< 0.1

SECOND PASS RO PERMEATE QUALITY REQUIREMENTS

PARAMETER (mg/l) UNLESS OTHERISE NOTED (U.O.N)	TYPICAL QUALITY 1 YEAR OPERATION	TYPICAL QUALITY 5 YEAR OPERATION	CONTRACTOR GUARANTEED QUALITY (mg/l) U.O.N
TDS	Contractor To Confirm	Contractor To Confirm	< 3
Conductivity (us/cm)	Contractor To Confirm	Contractor To Confirm	< 5
Chloride	Contractor To Confirm	Contractor To Confirm	< 1
Silica	Contractor To Confirm	Contractor To Confirm	< 0.05
pH	Contractor To Confirm	Contractor To Confirm	6.5 to 8
Calcium	Contractor To Confirm	Contractor To Confirm	< 0.2
Magnesium	Contractor To Confirm	Contractor To Confirm	< 0.1
Total Hardness	Contractor To Confirm	Contractor To Confirm	< 0.5
Fluoride	Contractor To Confirm	Contractor To Confirm	< 0.1
Sodium	Contractor To Confirm	Contractor To Confirm	< 1
Sulphate	Contractor To Confirm	Contractor To Confirm	< 0.05

FIRST PASS RO BRINE QUALITY REQUIREMENTS

PARAMETER (mg/l) UNLESS OTHERISE NOTED (U.O.N)	TYPICAL QUALITY 1 YEAR OPERATION	TYPICAL QUALITY 5 YEAR OPERATION	CONTRACTOR GUARANTEED QUALITY (mg/l) U.O.N
TDS	Contractor To Confirm	Contractor To Confirm	15834
Conductivity (us/cm)	Contractor To Confirm	Contractor To Confirm	Contractor To Confirm
Cl	Contractor To Confirm	Contractor To Confirm	< 932
SiO2	Contractor To Confirm	Contractor To Confirm	Contractor To Confirm
pH	Contractor To Confirm	Contractor To Confirm	6.5 to 7
Ca	Contractor To Confirm	Contractor To Confirm	453
Mg	Contractor To Confirm	Contractor To Confirm	282
Total Hardness	Contractor To Confirm	Contractor To Confirm	Contractor To Confirm
F	Contractor To Confirm	Contractor To Confirm	5.4
Na	Contractor To Confirm	Contractor To Confirm	4865
SO4	Contractor To Confirm	Contractor To Confirm	1881
K	Contractor To Confirm	Contractor To Confirm	272
Sr	Contractor To Confirm	Contractor To Confirm	4.04
Ba	Contractor To Confirm	Contractor To Confirm	1
CO3	Contractor To Confirm	Contractor To Confirm	0.14
HCO3	Contractor To Confirm	Contractor To Confirm	80.76
NO3	Contractor To Confirm	Contractor To Confirm	Contractor To Confirm
Boron	Contractor To Confirm	Contractor To Confirm	Contractor To Confirm
CO2	Contractor To Confirm	Contractor To Confirm	15.45

DEMINERALISED WATER QUALITY REQUIREMENTS

PARAMETER	GUARANTEED DEMIN WATER QUALITY	TYPICAL OPERATING CONCENTRATION
Ph	6.5 to 7.5	**Contractor to advise**
Conductivity µs/cm [1]	< 0.2	Contractor to advise
Iron ppb	< 10	Contractor to advise
Sodium ppb	< 5	Contractor to advise
Copper ppb	< 3	Contractor to advise
Total hardness meq/	< 0.003	Contractor to advise
Silica ppb	< 0.02	Contractor to advise
Dissolved Oxygen ppb [2]	< 6	Contractor to advise

POLISHED CONDENSATE WATER QUALITY REQUIREMENTS

PARAMETER	GUARANTEED DEMIN WATER QUALITY	TYPICAL OPERATING CONCENTRATION
Ph		Contractor to advise
Conductivity µs/cm [1]	< 0.2	Contractor to advise
Iron ppb	< 10	Contractor to advise
Sodium ppb	< 5	Contractor to advise
Copper ppb	< 3	Contractor to advise
Total hardness meq/	< 0.003	Contractor to advise
Silica ppb	< 10	Contractor to advise
Dissolved Oxygen ppb [2]	< 6	Contractor to advise

POTABLE WATER QUALITY REQUIREMENTS

Potable water quality shall meet all requirements of GCC STANDARDIZATION ORGANISATION (GSO) GSO5 / D5 /…/2012 (Unbottled Drinking water) and Royal Commission Directives.

WASTE DISCHARGE CONCENTRATION LIMIT FROM INDUSTRIAL WASTEWATER TREATMENT PLANT (IWTP) TO ENVIRONMENT

Wastewater treatment will be carried out in accordance with the following Specifications:

MD-101-SMEMEG-CV-SPC-0004 “Stormwater Drainage and Gravity Sewers”
MD-101-SMEM-EG-CV-SPC-0005 “Wastewater Treatment Plant”
MD-101-SMEM-EG-CV-SPC-0006 “Wastewater Treatment Plant”

DISCHARGE LIMITS OF WASTEWATER TO INDUSTRIAL WASTE WATER TREATMENT PLANT

PARAMETER	UNITS	Maximum Limit
PHYSICAL		
Temperature	°C	50
Total Dissolved Solids	mg/l	2500
Total Suspended Solids	mg/l	2000
CHEMICAL		
Aluminum	mg/l	30
Ammonia, Total as N	mg/l	120
Arsenic	mg/l	1.25
Barium	mg/l	2
BOD	mg/l	800
Boron	mg/l	2.5
Cadmium	mg/l	0.5
Chloride	mg/l	1000
Chlorinated hydrocarbons	mg/l	0.5
Chromium - Total	mg/l	5
Chromium - Hexavalent	mg/l	1
Cobalt	mg/l	2
COD	mg/l	1800
Copper	mg/l	1.2
Cyanide	mg/l	3.5
Fluoride	mg/l	150
Iron	mg/l	25
Lead	mg/l	0.5
Manganese	mg/l	2
Mercury	mg/l	0.015
Nickel	mg/l	2.5
Oil and Grease	mg/l	120
pH*	mg/l	5 - 9
Phenols	mg/l	150
Phosphorus, Total P	mg/l	50
Silver	mg/l	0.25
Sodium	mg/l	1000
Sulphate	mg/l	800
Sulphide	mg/l	10
Total Organic Carbon (TOC)	mg/l	800
Zinc	mg/l	10

REGULATORY STANDARDS FOR RESTRICTED IRRIGATION DISCHARGES

PARAMETER (1)	UNITS	Guide Value	Max Allowable	Max Allowable	Max Allowable	Monthly Average (30 day)
		General	(for reuse of water for irrigation - unoccupied lands / municipal areas not frequented by the public)	(for reuse of water for irrigation - municipal areas frequented by the public)	(for reuse of water for irrigation)	
		IFC / WB	MoWE (8)	MoWE (8)	RCER 2010	RCER 2010
PHYSICAL						
Floating Particles	mg/m2		Nil	Nil	Nil	
Total Suspended Solids	mg/l	50	40	10 (9)	10	10
Total Dissolved Solids	mg/l				2000	1750
Turbidity	N.T.U		5	5	5 (2)	2
CHEMICAL						
Aluminum	mg/l		5	5	5	
Ammonia, Total as N	mg/l		5	5	5	
Arsenic	mg/l		0.1	0.1	0.1	
Barium	mg/l				1	
Beryllium	mg/l		0.01	0.1	0.1	
BOD	mg/l	30	40	15 (9)	10 (4)	
Boron	mg/l		0.75	0.75	0.75	
Cadmium	mg/l		0.01	0.01	0.01	
COD	mg/l	125			50	
Chloride	mg/l				1000	500
Chlorine Residual	mg/l		0.5 (min)	0.5 (min)	0.5 (min) (3)	
Chromium	mg/l		0.1	0.1	0.01	
Cobalt	mg/l		0.05	0.05	0.05	
Copper	mg/l		0.4	0.4	0.2	
Cyanide	mg/l				0.05	
Dissolved Oxygen	mg/l				2.0 (min) (4)	
Fluoride	mg/l		1	1	15 (5)	5 (5)
Iron	mg/l		5	5	5	
Lead	mg/l		0.1	0.1	0.5	0.1
Lithium	mg/l		2.5	2.5	2.5	
Manganese	mg/l		0.2	0.2	0.2	0.02
Mercury	mg/l		0.001	0.001	0.001	
Molybdenum	mg/l		0.01	0.01	0.01	
Nickel	mg/l		0.2	0.2	0.02	
Nitrogen (total)	mg/l	10				
Nitrate	mg/l		10	10	10	
Oil and Grease	mg/l	10	Nil		Nil	
pH	pH units	6 - 9	6 - 8.4		6 - 8.4	
Phenols	mg/l		0.002	0.002	0.002	
Phosphorus, total as P	mg/l	2			30	20
Selenium	mg/l		0.02	0.02	0.02	
Silver -	mg/l				0.5	
Sodium	mg/l				1000	500
Sodium Adsorption Ratio (SAR)	SAR units				20	10
Sulfate -	mg/l				600	
Sulfide	mg/l				0.1	0.05
Total Kjeldahl Nitrogen	mg/l				60	35
Total Organic Carbon	mg/l				40	
Vanadium	mg/l		0.1	0.1	0.1	
Zinc	mg/l		4	4	2	
Total Chlorinated Hydro-carbons	mg/l					
BIOLOGICAL						
Total Coliform	MPN/100ml	400 (10)			23 (6)	2.2 (7)
Fecal Coliform	MPN/100ml		1000	2.2	1	
Nematodes (eggs)	No./ 100ml				1	
Protazoan Cysts	No./ 100ml				1	
Platyhelminths - flatworms	No./ 100ml			1live egg / l (11)	1	

Comments by SME on Civil Engineering Documents - by SME _ABDUL SHAKOOR_
MD-101-SMEM-EG-CV-SPC-0005_C02

#	Reference	Page	Description	Comments
01			General Comment	The subject specification is related to civil discipline but its content does contain anything about civil discipline. It largly discuss about mechanical equipment. Guidelines related civil structures need to be added.
02	3.2	5 of 25	Industry Codes & Standards	Industry codes and standards related civil structures can be included in the list.
03	6.1	15 of 25	Steel Work	Employer's specifications mentioned in this section are not included in the list of Maaden Standards referred in section 3.2.5. These specification can be included in that list also.
04	6	15 of 25	General Comment	Specifications for civil structures e.g. Concrete for equipment foundations, pedestals for pipe racks, trenches, buildings (technical, non technical) etc. Specifications for strength requirement, protection of concrete agains environment and chemicals etc. can be included in this section which are missing currently.